237 **Section 2, Module 1: Math**

[Meo] The function f is defined by $f(x) = \frac{12}{11-x}$. What is the value of $f(x)$ when $x = 9$?

6

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238 **Section 2, Module 1: Math**

[Meo] In $\triangle XYZ$, the measure of $\angle X$ is 20 degrees and the measure of $\angle Y$ is 87 degrees. What is the measure of $\angle Z$?

A 67 degrees

B 73 degrees

C 107 degrees

D 197 degrees

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239 **Section 2, Module 1: Math**

[Meo]

$$f(r) = 18 + 3r$$

The function f represents the total cost, in dollars, of attending a carnival when r rides are ridden. How many rides can be ridden for a total cost of 87 dollars?

23

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241 **Section 2, Module 1: Math**

[Meo] At a convention center, there are a total of 275 visitors. Each visitor is located in either room A, room B, or room C. If one of these visitors is selected at random, the probability of selecting a visitor who is located in room A is 0.44, and the probability of selecting a visitor who is located in room B is 0.16. How many visitors are located in room C?

A 19

B 40

C 77

D 110

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240 **Section 2, Module 1: Math**

[Meo] Which expression is equivalent to $5xy(2x^2 + 6y)$?

A $7x^3y + 11xy^2$

B $10x^2y + 6y$

C $10x^2y + 30xy$

D $10x^3y + 30xy^2$

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242 **Section 2, Module 1: Math**

[Meo] Which expression is a factor of $81p^{21} - 121p^{19}$?

A $9p + 11$

B $81p^2 + 121$

C $11p - 9$

D $-40p^{44}$

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244 **Section 2, Module 1: Math**

[Meo] Marko purchased a camera in August 2017 when the camera had a value of $1,000$ dollars. Each month for the next 50 months, the value of the camera decreased by 3% of the previous month's value. Which equation represents this situation, where v is the value of the camera, in dollars, m is the number of months since Marko purchased the camera, and $m \leq 50$?

A $v = 2,017(0.97)^m$

B $v = 2,017(3)^m$

C $v = 1,000(0.97)^m$

D $v = 1,000(3)^m$

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243 **Section 2, Module 1: Math**

[Meo] A cube has an edge length of 36 inches. What is the volume, in cubic inches, of the cube?

A $46,656$

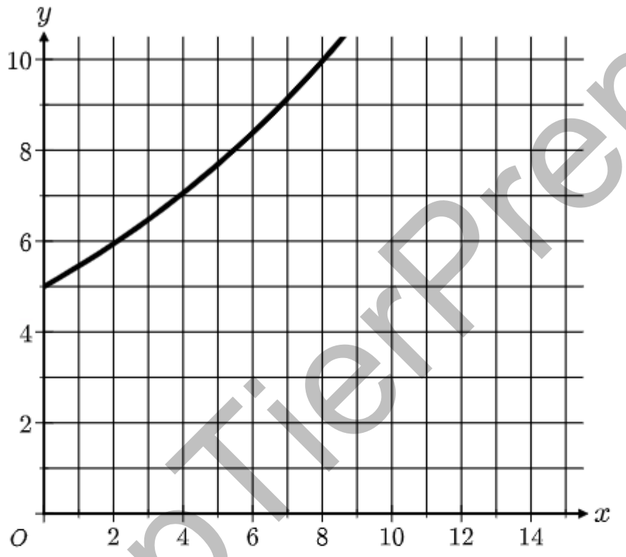
B $7,776$

C $1,296$

D 144

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245 **Section 2, Module 1: Math**



[Meo] The graph models the population, in thousands, of a town x years since 2003, where $0 \leq x \leq 15$. Which statement is the best interpretation of the point $(1, 5.45)$?

A In 2003, the estimated population of the town was $5,450$.

B In 2004, the estimated population of the town was $5,450$.

C In 2017, the estimated population of the town was $5,450$.

D In 2018, the estimated population of the town was $5,450$.

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246 Section 2, Module 1: Math

[Meo]

$$p(x) = 131 - 3x$$

The function p shown gives the number of blank sheets of paper, $p(x)$, remaining in a printer x days after the printer had been loaded with paper. Which statement is the best interpretation of $p(5) = 116$?

A The number of blank sheets of paper remaining in the printer decreased by a total of 5 sheets after 116 days.

B There were 5 blank sheets of paper remaining in the printer 116 days after the printer had been loaded with paper.

C There were 116 blank sheets of paper remaining in the printer 5 days after the printer had been loaded with paper.

D The number of blank sheets of paper remaining in the printer decreased by a total of 116 sheets after 5 days.

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247 **Section 2, Module 1: Math**

[Meo]

$$y = (x - 7)(x + 6)$$

$$y = 4x - 28$$

A solution to the given system of equations is $(x, y) = (a, b)$. Which of the following could be the value of a?

A 7

B 6

C -6

D -7

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248 **Section 2, Module 1: Math**

[Meo] Rattlesnakes do not lay eggs but give birth to live young. For the last three years, a rattlesnake at a zoo gave birth to one litter per year. The table shows the number of rattlesnakes in each of these three litters.

Litter number	Number of rattlesnakes
1	13
2	11 **Meo**
3	15 **Meo**

What is the mean number of rattlesnakes per litter for these three litters?

A 2

B 11

C 13

D 15

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249 **Section 2, Module 1: Math**

[Meo] Hassan put up wire fencing along each edge of a rectangular garden with a length of x feet and a width of y feet. Hassan put up a total length of 22 feet of wire fencing. Which equation represents this situation?

A $xy = 22$

B $2xy = 22$

C $x + y = 22$

D $2x + 2y = 22$

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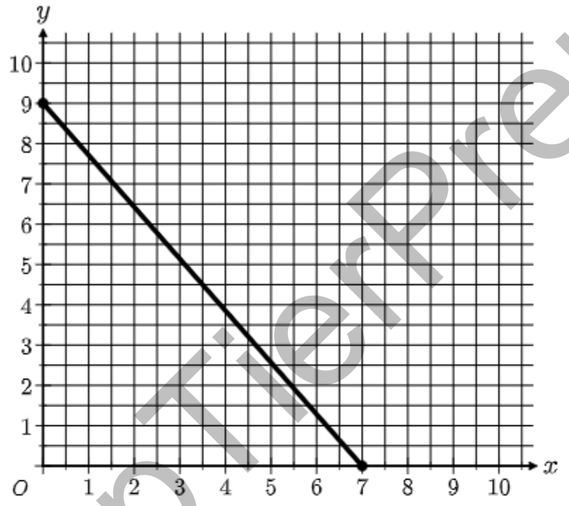
250 **Section 2, Module 1: Math**

[Meo] The price of an item increased by $p\%$ from \$78 to \$81. What is the value of p?

50/13 3.846

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251 **Section 2, Module 1: Math**



[Meo] A line segment is shown in the xy-plane. If line ℓ (not shown) is perpendicular to the line segment shown and passes through the point $(0, 6)$, which equation defines line ℓ?

(A) $9x + 7y = 42$

(B) $-9x + 7y = 42$

(C) $7x + 9y = 54$

(D) $-7x + 9y = 54$

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252 **Section 2, Module 1: Math**

[Meo] The exponential function g is defined by $g(x) = ab^x$, where a and b are positive constants. If $g(c) = m$ and the value of $g(c + 1)$ is 75% more than the value of m, where c and m are constants, what is the value of b?

A 0.25

B 0.75

C 1.75

D 3.5

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253 **Section 2, Module 1: Math**

[Meo]

$$(2x^3 - 8) - (-5x^3 + 7)$$

The given expression is equivalent to $bx^3 - 15$, where b is a constant. What is the value of b?

7

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254 **Section 2, Module 1: Math**

[Meo] A circle in the xy-plane has its center at $(-9, -8)$ and intersects the line $y = a$ at exactly one point, where a is a positive constant. Which equation defines this circle?

A $(x+9)^2 + (y+8)^2 = (a+8)^2$

B $(x+9)^2 + (y+8)^2 = (a-8)^2$

C $(x-9)^2 + (y-8)^2 = (a+8)^2$

D $(x-9)^2 + (y-8)^2 = (a-8)^2$

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255 **Section 2, Module 1: Math**

[Meo] A machine makes 3-inch, 7-inch, and 10-inch parts. During a certain day, the number of 3-inch parts that the machine makes is 5 times the number n of 7-inch parts, and the number of 10-inch parts is 4. During this day, the machine makes 100 parts total. Which equation represents this situation?

A $3(5n) + 7n + 10(4) = 100$

B $3n + 7n + 10n = 100$

C $5n + 4 = 100$

D $6n + 4 = 100$

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256 **Section 2, Module 1: Math**

[Meo]

$$44x^2 + (44b + a)x + ab = 0$$

In the given equation, a and b are positive constants. The product of the solutions to the given equation is kab, where k is a constant. What is the value of k?

A $\frac{1}{44}$

B $\frac{1}{11}$

C 1

D 44

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257 **Section 2, Module 1: Math**

[Meo] The function f is defined by $f(x) = 65x - 40$. What is the value of $f(x)$ when $x = 8$?

480

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258 **Section 2, Module 1: Math**

[Meo] In triangle XYZ, two of the angles have the same measure. If the measure of angle X is 158 degrees and the measure of angle Y is 11 degrees, what is the measure of angle Z?

A 180 degrees

B 158 degrees

C 22 degrees

D 11 degrees

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260 **Section 2, Module 1: Math**

[Meo] The function g is defined by $g(x) = \sqrt{7x + 4}$. What is the value of $g(3)$?

5

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259 **Section 2, Module 1: Math**

[Meo]

$$\sqrt{w} + 35 = 45$$

What is the solution to the given equation?

A 100

B 20

C 5

D 2

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261 **Section 2, Module 1: Math**

[Meo] The equation $56 = 2x + 2y$ gives the perimeter of a rectangular garden that has length x, in feet, and width y, in feet. The width of the garden is 13 feet. What is the length, in feet, of the garden?

15

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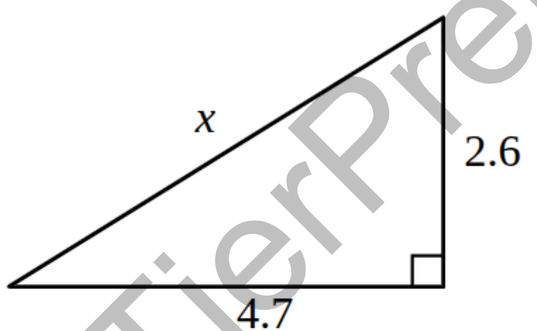
262 **Section 2, Module 1: Math**

[Meo] The ratio of the side length of square A to the side length of square B is 4 to 20. Square A has a side length of 11 inches. What is the side length, in inches, of square B?

55

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263 **Section 2, Module 1: Math**



Note: Figure not drawn to scale.

[Meo] For the right triangle shown with legs of length 2.6 and 4.7 and hypotenuse of length x, the value of x is between which two consecutive integers?

A 2 and 3

B 5 and 6

C 7 and 8

D 28 and 29

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264 **Section 2, Module 1: Math**

[Meo] The measure of angle Z is 15 degrees. What is the measure, in radians, of angle Z?

A $\frac{1}{24}\pi$

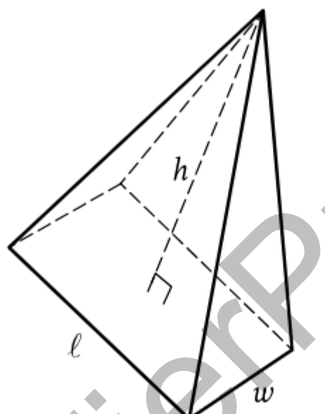
B $\frac{1}{12}\pi$

C $\frac{1}{6}\pi$

D $\frac{1}{4}\pi$

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265 **Section 2, Module 1: Math**



Note: Figure not drawn to scale.

[Meo] The figure shown is a right rectangular pyramid, where $\ell = 10$ units, $w = 5$ units, and $h = 11$ units. What is the volume, in cubic units, of the pyramid?

550/3

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266 **Section 2, Module 1: Math**

[Meo]

$$(7x^3 - 3) - (-6x^3 + 5)$$

The given expression is equivalent to $bx^3 - 8$, where b is a constant. What is the value of b?

13

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267 **Section 2, Module 1: Math**

[Meo]

$$6(x-3)=13+4(x-3)$$

If x is the solution to the given equation, what is the value of $x-3$?

13/2 6.5

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268 **Section 2, Module 1: Math**

[Meo] Line k is defined by $y = 7x + \frac{1}{4}$. Line j is perpendicular to line k in the xy-plane. What is the slope of line j?

A -4

B $-\frac{1}{7}$

C $\frac{1}{4}$

D 7

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269 **Section 2, Module 1: Math**

[Meo] The graph of a polynomial function $y = f(x)$ in the xy-plane passes through the point $(9, 5)$. What is the value of $f(9)$?

A 5

B 9

C 14

D 45

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271 Section 2, Module 1: Math

[Meo]

$$x(x+11)-15=0$$

What are the solutions to the given equation?

A $x=\frac{11\pm\sqrt{61}}{2}$

B $x=\frac{11\pm\sqrt{181}}{2}$

C $x=\frac{-11\pm\sqrt{61}}{2}$

D $x=\frac{-11\pm\sqrt{181}}{2}$

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270 **Section 2, Module 1: Math**

[Meo] The function f is defined by $f(x) = 4x - 6$. If $f(a) + 1 = 3a$, what is the value of a?

A 7

B 5

C 1

D -1

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272 **Section 2, Module 1: Math**

[Meo]

$$g(x) = x^2 + 6x + 90$$

The function g is defined by the given equation. Which of the following equivalent forms of the equation displays the minimum value of the function as a constant or coefficient?

A $g(x) = (x + 3)^2 + 81$

B $g(x) = x^2 + 2(3x + 45)$

C $g(x) - 90 = x(x + 6)$

D $g(x) - 90 = \frac{1}{2}x(2x + 12)$

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273 Section 2, Module 1: Math

[Meo]

x	y
-6	$n + 142$
-3	$n + 71$
0	n **Meo**

The table shows three values of x and their corresponding values of y, where n is a constant, for the linear relationship between x and y. What is the slope of the line that represents this relationship in the xy-plane?

A $-\frac{71}{3}$

B $-\frac{3}{71}$

C $\frac{n+71}{-3}$

D $\frac{2n-71}{3}$

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274 **Section 2, Module 1: Math**

[Meo] During a trip, a car traveled a total of 450 miles on city roads and highways and used a total of 15 gallons of gas. The car traveled an average of 29 miles per gallon of gas on city roads and 35 miles per gallon of gas on highways. Which system of equations represents this situation, where c is the number of miles the car traveled on city roads and h is the number of miles the car traveled on highways?

A $c + h = 15$
$\frac{c}{29} + \frac{h}{35} = 450$

B $c + h = 15$
$29c + 35h = 450$

C $c + h = 450$
$\frac{c}{29} + \frac{h}{35} = 15$

D $c + h = 450$
$29c + 35h = 15$

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275 **Section 2, Module 1: Math**

[Meo] The exponential function g is defined by $g(x) = ab^x$, where a and b are positive constants. If $g(c) = m$ and the value of $g(c + 1)$ is 68% more than the value of m, where c and m are constants, what is the value of b?

(A) 0.32

(B) 0.68

(C) 1.68

(D) 3.36

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276 **Section 2, Module 1: Math**

[Meo] Cylinder A and cylinder B are each right circular cylinders with a radius of 4 inches. The height of cylinder A is half the height of cylinder B, and the total surface area of cylinder A is 128π square inches. If the cylinders are glued together along a circular base, what is the height, in inches, of the resulting cylinder?

A 8

B 12

C 24

D 36

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277 **Section 2, Module 1: Math**

[Meo]

Meo	Live north of Center St. **Meo**	Live south of Center St.	Total
Less than 45 years old **Meo**	11	20 **Meo**	31
At least 45 years old	24 **Meo**	80 **Meo**	104 **Meo**
Total	35	100 **Meo**	135

The table summarizes members of a local organization by age and whether they live north or south of Center St. If a member of the organization is selected at random, what is the probability that the selected member is at least 45 years old?

A $\frac{31}{135}$

B $\frac{35}{135}$

C $\frac{100}{135}$

D $\frac{104}{135}$

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278 **Section 2, Module 1: Math**

[Meo] The ratio of the side length of square A to the side length of square B is 4 to 20. Square A has a side length of 22 inches. What is the side length, in inches, of square B?

110

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279 **Section 2, Module 1: Math**

[Meo]

$$(x - 10)(x + 3) = 0$$

How many distinct real solutions does the given equation have?

A Zero

B Exactly one

C Exactly two

D Infinitely many

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280 **Section 2, Module 1: Math**

[Meo] Which expression is a factor of $81p^{21} - 121p^{19}$?

A $9p + 11$

B $81p^2 + 121$

C $11p - 9$

D $-40p^{44}$

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281 **Section 2, Module 1: Math**

[Meo] The estimated fuel cost $C(x)$, in dollars, to drive a vehicle 100 kilometers is given by the function $C(x) = 0.0021x^2 - 0.278x + 14.14$, where x is the average speed, in kilometers per hour, of the vehicle. If the function C is graphed in the xy-plane, where $y = C(x)$, what is the best interpretation of the point $(60, 5.02)$ on the graph?

A The estimated fuel cost to drive the vehicle at an average speed of 5.02 kilometers per hour increases by 60 dollars each hour.

B The estimated fuel cost to drive the vehicle at an average speed of 60 kilometers per hour increases by 5.02 dollars each hour.

C The estimated fuel cost to drive the vehicle 100 kilometers at an average speed of 5.02 kilometers per hour is 60 dollars.

D The estimated fuel cost to drive the vehicle 100 kilometers at an average speed of 60 kilometers per hour is 5.02 dollars.

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282 **Section 2, Module 1: Math**

[Meo] Two types of fish will be added to a pond: bluegill and largemouth bass. Based on ecological guidelines, the number of bluegill added to the pond will be 10 times the number of largemouth bass added to the pond. If a total of 3,410 fish will be added to the pond, how many of these will be largemouth bass?

310

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283 **Section 2, Module 1: Math**

[Meo]

$$4x^2 + x - 4 = 0$$

What is the greatest solution to the given equation?

A $-\frac{1}{8} - \frac{\sqrt{65}}{8}$

B $-\frac{1}{8} + \frac{\sqrt{65}}{8}$

C $-\frac{1}{4} - \frac{\sqrt{65}}{4}$

D $-\frac{1}{4} + \frac{\sqrt{65}}{4}$

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284 **Section 2, Module 1: Math**

[Meo]

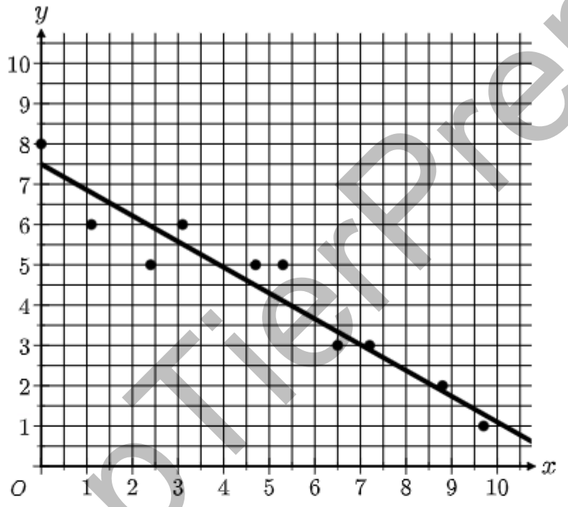
$$f(x) = (x - a)(x - b)$$

The function f is defined by the given equation, where a and b are integer constants. If $f(11) > 0$, $f(14) < 0$, and $f(17) > 0$, what is one possible value of $a + b$?

27 **28** **29**

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285 Section 2, Module 1: Math



[Meo] In the given scatterplot, a line of best fit for the data is shown, passing approximately through the points $(0, 7.5)$ and $(10, 1.5)$. Which of the following is closest to the slope of this line of best fit?

A -6

B -0.6

C 0.6

D 6

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286 **Section 2, Module 1: Math**

[Meo] In the xy-plane, the graph of the equation $(x-6)^2+(y-7)^2=36$ is a circle. The point $(12, c)$, where c is a constant, lies on this circle. What is the value of c?

7

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287 **Section 2, Module 1: Math**

[Meo] The length of a rectangle's diagonal is $\sqrt{106}$, and the length of one of the rectangle's sides is 9. What is the perimeter of the rectangle?

A 106

B 45

C 28

D 14

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288 **Section 2, Module 1: Math**

[Meo] How many solutions does the equation $17x = 17x$ have?

(A) Zero

(B) Exactly one

(C) Exactly two

(D) Infinitely many

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289 **Section 2, Module 1: Math**

[Meo] To install flooring, a contractor charges 5.75 dollars per square foot plus 100 dollars for labor. If a kitchen floor is a square with the side length of 17 feet, what is the total, in dollars, that the contractor charges to install flooring in this kitchen?

A 197.75

B 394.75

C 1,705.75

D 1,761.75

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290 **Section 2, Module 1: Math**

[Meo] In the relationship between x and y, each increase of 1 in the value of x decreases the value of y by $\frac{6}{7}$. When $x = -7$, $y = -1$. Which equation represents this relationship?

A $y = -\frac{6}{7}x - 7$

B $y = -\frac{6}{7}x + 7$

C $y = \frac{6}{7}x - 7$

D $y = \frac{6}{7}x + 7$

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291 **Section 2, Module 1: Math**

[Meo]

$$20(0.6r - 0.8s) = -1,840$$

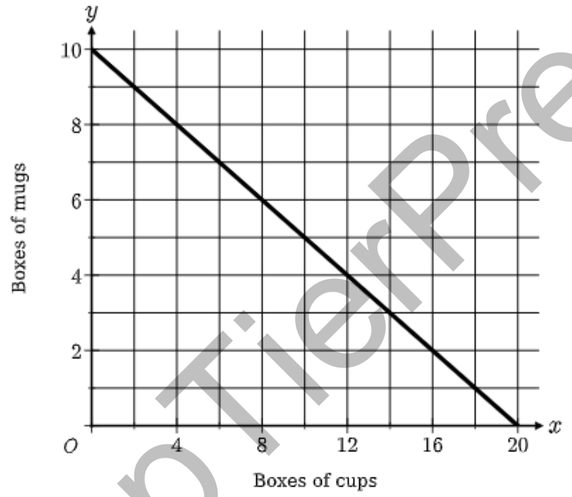
$$70(0.8r + 0.6s) = 6,580$$

The solution to the given system of equations is (r, s). What is the value of $r + s$?

150

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292 **Section 2, Module 1: Math**



[Meo] The line models the possible combinations of the number of boxes of cups, x, and the number of boxes of mugs, y, a company can purchase for a total price of 980 dollars. What is the price, in dollars, per box of cups?

A 10

B 20

C 49

D 98

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293 **Section 2, Module 1: Math**

[Meo] When several resistors are connected in series in a circuit, the total resistance of the circuit is the sum of the resistances of these resistors. A certain circuit consists of 9 resistors connected in series, where the resistance of each resistor is positive. The total resistance of 4 resistors in this circuit is 50 ohms. Each of the other 5 resistors in this circuit has a resistance of at most 20 ohms. The total resistance of this circuit is x ohms. Which inequality best represents this situation?

A $0 < x \leq 150$

B $50 < x \leq 100$

C $50 < x \leq 150$

D $x \geq 100$

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294 **Section 2, Module 1: Math**

[Meo]

$$f(x) = 7((\frac{1}{6})^x + 4)$$

The function f is defined by the given equation. For all values of x, the value of $f(x)$ is greater than k, where k is a constant. What is the greatest possible value of k?

A 0

B 4

C 28

D 168

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295 **Section 2, Module 1: Math**

[Meo]

x	y **Meo**
-5 **Meo**	$49+4k$ **Meo**
5	49
10	$49-2k$

The table gives three values of x and their corresponding values of y, where k is a positive constant. There is a linear relationship between x and y. Which equation represents this relationship?

A $2kx-5y=49+2k$

B $2kx-5y=245+10k$

C $2kx+5y=49+2k$

D $2kx+5y=245+10k$

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296 **Section 2, Module 1: Math**

[Meo]

$$x - a = (x - a)(x - 27)$$

Which of the following are solutions to the given equation, where a is a constant and $a > 28$?

I. a
II. 27
III. 28

A I and II only

B I and III only

C II and III only

D I, II, and III

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297 **Section 2, Module 1: Math**

[Meo] An entomologist is studying the growth rate of various insect species in a controlled environment. For one particular species being studied in the controlled environment, an exponential model predicts that a population with an initial measurement of 10 organisms will increase by 160% every 22 days. Which equation represents this model, where y is the predicted number of organisms in the population x days since the initial measurement?

A $y = 10(1.60)^{22x}$

B $y = 10(2.60)^{22x}$

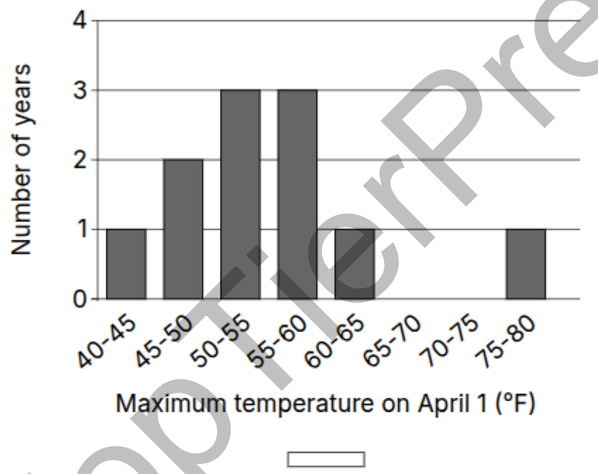
C $y = 10(1.60)^{\frac{x}{22}}$

D $y = 10(2.60)^{\frac{x}{22}}$

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298 **Section 2, Module 1: Math**

[Meo] The maximum temperature on April 1, in degrees Fahrenheit, was recorded each year at a certain weather station for 11 years. The histogram summarizes the recorded data set.



The temperature of 77.4 degrees Fahrenheit is removed from this data set to create a new data set of 10 temperatures. Which of the following statements must be true?

I. The mean of the new data set is less than the mean of the original data set.
II. The median of the new data set is less than the median of the original data set.

A I only

B II only

C I and II

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299 **Section 2, Module 1: Math**

[Meo] The positive number a is $2,047\%$ of the sum of the positive numbers b and c, and b is 89% of c. What percent of b is a?

A $4,347\%$

B $2,136\%$

C 43.47%

D 21.36%

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300 **Section 2, Module 1: Math**

[Meo] A line in the xy-plane passes through the points $(9, 1)$, $(0, 8)$, and $(c, 0)$. What is the value of c?

72/7

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301 **Section 2, Module 1: Math**

[Meo] A pet service that takes care of both cats and dogs can care for at most 41 animals. Which inequality represents this situation, where c is the number of cats and d is the number of dogs?

A $c + d \leq 41$

B $c + d \geq 41$

C $c - d \leq 41$

D $c - d \geq 41$

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302 **Section 2, Module 1: Math**

[Meo] The length of a rectangle is 41 centimeters greater than its width. Which equation gives the area A, in square centimeters, of the rectangle in terms of the width w, in centimeters, of the rectangle?

A $A = (w)(w + 41)$

B $A = (w)(w - 41)$

C $A = (w)(41w)$

D $A = (w)(\frac{w}{41})$

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303 **Section 2, Module 1: Math**

[Meo] In each of the following data sets of 5 values, p is a constant. Which of these data sets has the largest standard deviation?

A $p-1, p-1, p, p+1, p+1$

B $p-2, p, p, p, p+2$

C p, p, p, p, p

D $p-4, p-2, p, p+2, p+4$

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304 **Section 2, Module 1: Math**

[Meo] The radius of circle P is 210 centimeters. What is the diameter of circle P, in centimeters?

420

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305 **Section 2, Module 1: Math**

[Meo] The function $P(x) = 685(1.128)^x$ models the number of gray seal pups each year from 1967 through 1997 on Sable Island, Nova Scotia, where x is the number of years after 1967. Which of the following is the best interpretation of \"$P(16)$ is approximately equal to $4,706$\" in this context?

A The number of gray seal pups is estimated to be approximately $4,706$ in 1983.

B The number of gray seal pups is estimated to be 16 greater in 1983 than in 1967.

C The number of gray seal pups is estimated to be 16 times greater in 1983 than in 1967.

D The number of gray seal pups is estimated to increase by approximately $4,706$ every 16 years between 1967 and 1983.

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306 **Section 2, Module 1: Math**

[Meo] Sam opened a savings account at a bank. The table

Time (years) **Meo**	Total amount (dollars)
0 **Meo**	864.00
1	869.18
2 **Meo**	874.40

shows the exponential relationship between the time t, in years, since Sam opened the account and the total amount a, in dollars, in the account. If Sam made no additional deposits or withdrawals, which of the following equations best represents the relationship between t and a?

A $a = 0.006(1 + 864)^t$

B $a = 864(1 + 0.006)^t$

C $a = (1 + 0.006)^t$

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307 **Section 2, Module 1: Math**

[Meo] A linear function g is defined by $g(x) = ax + b$, where a and b are constants. If $g(-2) = -7$ and $g(1) = 5$, what is the value of a?

4

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308 **Section 2, Module 1: Math**

[Meo] Hassan put up wire fencing along each edge of a rectangular garden with a length of x feet and a width of y feet. Hassan put up a total length of 28 feet of wire fencing. Which equation represents this situation?

A $2x + 2y = 28$

B $x + y = 28$

C $2xy = 28$

D $xy = 28$

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309 **Section 2, Module 1: Math**

[Meo]

$$7x^2(4x - 11)(4x - u) = 0$$

In the given equation, u is a positive constant. The sum of the solutions to the equation is 9. What is the value of u?

A 9

B 11

C 25

D 52

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310 **Section 2, Module 1: Math**

[Meo] Two sets of blocks include both wood-finish and painted blocks. The tablesummarizes the types of blocks in each set.

Set	Wood-finish blocks	Painted blocks	Total
A	9 **Meo**	8	17 **Meo**
B **Meo**	11 **Meo**	3	14
Total	20	11 **Meo**	31 **Meo**

If one of these blocks is selected at random, what is the probability of selecting a wood-finish block that is in set A? (Express your answer as a decimal or fraction, not as a percent.)

9/31

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311 **Section 2, Module 1: Math**

[Meo]

$$y \geq x + 6$$

$$y \leq -3x - 6$$

Which point (x, y) is a solution to the given system of inequalities in the xy-plane?

A $(-4, 5)$

B $(-4, 1)$

C $(-2, 5)$

D $(-2, 1)$

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312 **Section 2, Module 1: Math**

[Meo] What is an x-coordinate of an x-intercept of the graph of $y = 3(x - 13)(x + 5)(x + 1)$ in the xy-plane?

13 -5 -1

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313 **Section 2, Module 1: Math**

[Meo]

$$v = -\frac{w}{133x}$$

The given equation relates the distinct numbers v, w, and x. Which equation correctly expresses w in terms of v and x?

A $w = -133vx$

B $w = -\frac{133v}{x}$

C $w = -\frac{v}{133x}$

D $w = v + 133x$

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314 **Section 2, Module 1: Math**

[Meo]

$$y = 4x - 13$$

$$7x + y = 42$$

The solution to the given system of equations is (x, y). What is the value of $x + y$?

12

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315 **Section 2, Module 1: Math**

[Meo]

$$y = 9x^2 - 54x + 45$$

The given equation represents a parabola in the xy-plane. Which of the following equations that represent the same parabola displays the x-intercepts as constants or coefficients?

A $y = 9(x - 1)(x - 5)$

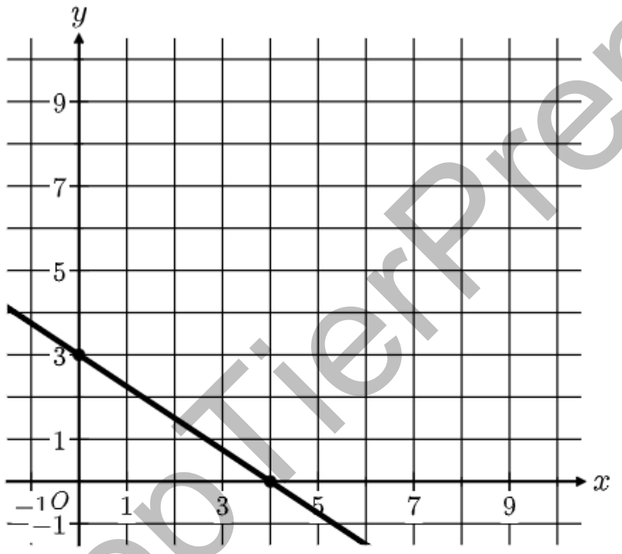
B $y = 9(x - 3)^2 - 36$

C $y = 9x(x - 6) + 45$

D $y = 9(x^2 - 6x) + 45$

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316 **Section 2, Module 1: Math**



[Meo] The graph of the linear equation $6x + By = 24$ is shown, where B is a constant. What is the value of B?

A $\frac{3}{4}$

B 3

C 4

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317 **Section 2, Module 1: Math**

[Meo] Triangle ABC is similar to triangle DEF, where angle A corresponds to angle D and angle C corresponds to angle F. Angles C and F are right angles. If $tan(A) = \frac{50}{27}$, what is the value of $\tan(E)$?

27/50

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318 **Section 2, Module 1: Math**

[Meo] Data set A consists of 10 positive integers less than 60. The list gives 9 of the integers from data set A.

$$40, 43, 42, 40, 36, 37, 39, 44, 39$$

The mean of these 9 integers is 40. If the mean of data set A is an integer that is greater than 40, what is the value of the largest integer from data set A?

A) 59

B) 50

C) 44

D) 41

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319 **Section 2, Module 1: Math**

[Meo] An online store is offering two different discounts on its sweaters: a bulk discount for large orders and a coupon discount. For orders of n sweaters, where $10 < n < 50$, a bulk discount of $n\%$ off the original price of 39 dollars is applied to the price of each sweater. If a coupon is applied to the order, a discount of an additional 3 dollars is applied to the price of each sweater, after the bulk discount is applied. For a certain order with both the bulk discount and the coupon discount applied, the total price after the discounts for the sweaters purchased was 729 dollars. How many sweaters were purchased in this order?

A 19

B 20

C 30

D 44

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320 **Section 2, Module 1: Math**

[Meo] An entomologist is studying the growth rate of various insect species in a controlled environment. For one particular species being studied in the controlled environment, an exponential model predicts that a population with an initial measurement of 8 organisms will increase by 160% every 22 days. Which equation represents this model, where y is the predicted number of organisms in the population x days since the initial measurement?

A $y = 8(1.60)^{22x}$

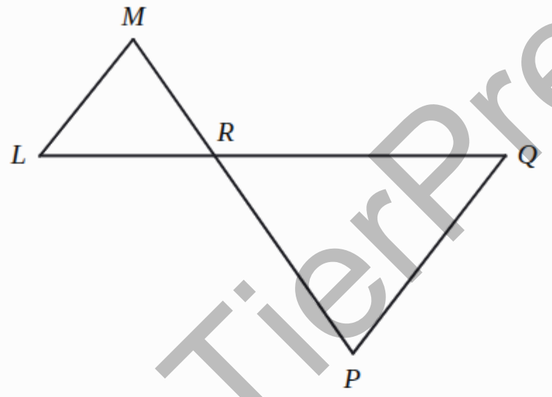
B $y = 8(2.60)^{22x}$

C $y = 8(1.60)^{\frac{x}{22}}$

D $y = 8(2.60)^{\frac{x}{22}}$

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321 **Section 2, Module 1: Math**



Note: Figure not drawn to scale.

[Meo] In the figure, $\overline{LQ}$ intersects $\overline{MP}$ at point R, and $\overline{LM}$ is parallel to $\overline{PQ}$. The lengths of $\overline{MR}$, $\overline{LR}$, and $\overline{RP}$ are 9, 10, and 11, respectively. What is the length of $\overline{LQ}$?

A $\frac{110}{9}$

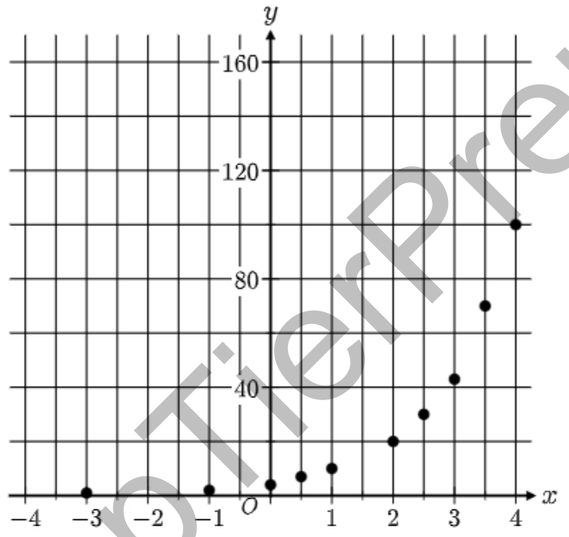
B $\frac{200}{11}$

C $\frac{191}{9}$

D 200

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322 **Section 2, Module 1: Math**



[Meo] The scatterplot shows the relationship between x and y for the 10 data points in data set A. Data set B is created by multiplying the y-value of each data point from data set A by 30. Which of the following equations is the most appropriate model for data set B?

A $y = 5(2.04)^x$

B $y = 149(2.04)^x$

C $y = 5(2.04)^{30}$

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323 **Section 2, Module 1: Math**

[Meo]

$$34x^2 + (34b + a)x + ab = 0$$

In the given equation, a and b are positive constants. The product of the solutions to the given equation is kab, where k is a constant. What is the value of k?

A $\frac{1}{34}$

B $\frac{1}{17}$

C 1

D 34

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324 **Section 2, Module 1: Math**

[Meo] A cube has an edge length of 31 inches. What is the volume, in cubic inches, of the cube?

A 124

B 961

C 5,766

D 29,791

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325 **Section 2, Module 1: Math**

[Meo] Marko purchased a camera in September 2017 when the camera had a value of 1,000 dollars. Each month for the next 50 months, the value of the camera decreased by 3% of the previous month's value. Which equation represents this situation, where v is the value of the camera, in dollars, m is the number of months since Marko purchased the camera, and $m \leq 50$?

A $v = 2,017(0.97)^m$

B $v = 2,017(3)^m$

C $v = 1,000(0.97)^m$

D $v = 1,000(3)^m$

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326 **Section 2, Module 1: Math**

[Meo]

$$y = (x - 8)(x + 5)$$

$$y = 2x - 16$$

A solution to the given system of equations is $(x, y) = (a, b)$. Which of the following could be the value of a?

A -8

B -5

C 5

D 8

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327 **Section 2, Module 1: Math**

[Meo]

x	$f(x)$ **Meo**
1	7,000
2	2,800
3 **Meo**	1,120
4	448

For the function f, the table shows four values of x and their corresponding values of $f(x)$. Which of the following could describe this function?

A Increasing linear

B Decreasing linear

C Increasing exponential

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328 **Section 2, Module 1: Math**

[Meo] Which of the following is equivalent to $\frac{44(k+2)}{4(k+2)}$, where $k > -2$?

A 11

B 13

C $13k$

D $11k + 2$

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329 **Section 2, Module 1: Math**

[Meo] Line k is defined by $y = 6x + \frac{1}{7}$. Line j is perpendicular to line k in the xy-plane. What is the slope of line j?

A -7

B $-\frac{1}{6}$

C $\frac{1}{7}$

D 6

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330 **Section 2, Module 1: Math**

[Meo]

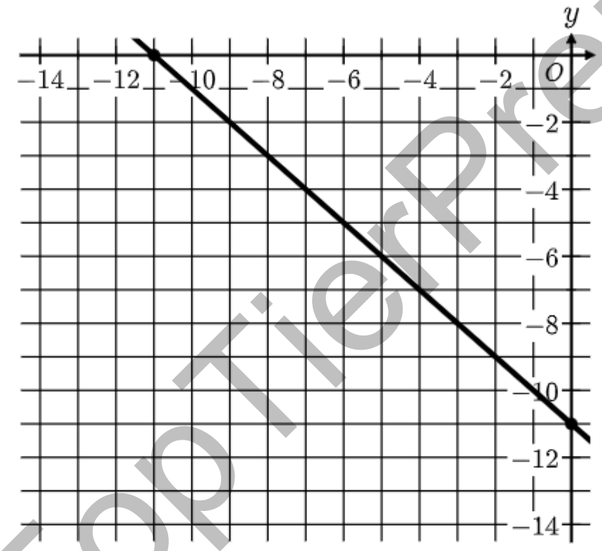
$$\frac{1}{x-1} = \frac{x-4}{-x+0.25}$$

What is one of the solutions to the given equation?

1.5	3/2	2.5	5/2

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331 **Section 2, Module 1: Math**



[Meo] The point $(v, -3)$ lies on this line. What is the value of v?

A $-\frac{96}{11}$

B $-\frac{11}{12}$

C $\frac{96}{11}$

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332 Section 2, Module 1: Math

[Meo]

Time (years)	Total amount (dollars) **Meo**
0 **Meo**	959.00 **Meo**
1	963.80
2 **Meo**	968.61

Sam opened a savings account at a bank. The table shows the exponential relationship between the time t, in years, since Sam opened the account and the total amount n, in dollars, in the account. If Sam made no additional deposits or withdrawals, which of the following equations best represents the relationship between t and n?

A $n = 0.005(1 + 959)^t$

B $n = 959(1 + 0.005)^t$

C $n = (1 + 0.005)^t$

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333 **Section 2, Module 1: Math**

[Meo] A total of 2 equilateral pentagons each have side length p units. A total of 6 squares each have side length s units. The sum of the perimeters of all these pentagons and squares is 112 units. Which equation represents this situation?

A $8p + 30s = 112$

B $6p + 2s = 112$

C $2p + 6s = 112$

D $10p + 24s = 112$

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334 **Section 2, Module 1: Math**

[Meo] The Southern Oscillation Index (SOI) predicts weather patterns in the Pacific Ocean. SOI values less than -7 over a period of time predict a temporary decrease in the strength of the Pacific trade winds. Each of the following inequalities represents SOI values, s, over a certain period of time. Which inequality represents SOI values that predict a temporary decrease in the strength of the Pacific trade winds?

A $-27 < s < -17$

B $-7 < s < 0$

C $-7 < s < 7$

D $19 < s < 24$

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335 **Section 2, Module 1: Math**

[Meo] The function f is defined by $f(x) = (x - 5) + 1$. If $f(a) = 7$, what is the value of $a + 5$?

(A) 35

(B) 16

(C) 12

(D) 6

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336 **Section 2, Module 1: Math**

[Meo] The expression $7(9x + 13)(x - 3) - (5x + 19)$ can be written in the form $ax^2 + bx + c$, where a, b, and c are constants. What is the value of b?

-103

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337 **Section 2, Module 1: Math**

[Meo]

$$y > 3x + k$$

$$y < 7 - x$$

In the given system of inequalities, k is a positive constant. Which of the following represents the x-coordinates of all the points (x, y) that satisfy this system in the xy-plane?

A $x > \frac{k}{7}$

B $x < \frac{k}{7}$

C $x > \frac{7-k}{4}$

D $x < \frac{7-k}{4}$

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338 **Section 2, Module 1: Math**

[Meo]

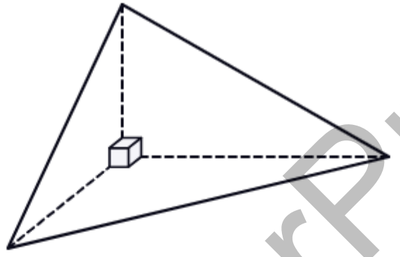
$$f(x) = (x - a)(x - b)$$

The function f is defined by the given equation, where a and b are integer constants. If $f(25) > 0$, $f(28) < 0$, and $f(31) > 0$, what is one possible value of $a + b$?

55 **56** **57**

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339 Section 2, Module 1: Math



Note: Figure not drawn to scale.

[Meo] The figure shown is a trirectangular tetrahedron, a three-dimensional figure with four triangular faces, three of which are right triangles. A chemistry professor uses trirectangular tetrahedrons to create large-scale models of various carbon-based molecules. The base of one of these tetrahedrons is an isosceles right triangle whose legs each have length 12 inches (in). The height of this tetrahedron is 24 in. Which of the following is closest to the volume of this tetrahedron, in pints? (Note: 1 pint = 0.5 quarts; use 1 quart = 57.75 in^3. The volume of a trirectangular tetrahedron is equal to $\frac{1}{3}Sh$, where S is the area of its base and h is its height.)

A 19.95

B 119.69

C 576.00

D 16,632.00

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340 **Section 2, Module 1: Math**

[Meo] An online store is offering two different discounts on its sweaters: a bulk discount for large orders and a coupon discount. For orders of n sweaters, where $10 < n < 50$, a bulk discount of $n\%$ off the original price of 32 dollars is applied to the price of each sweater. If a coupon is applied to the order, a discount of an additional 2 dollars is applied to the price of each sweater, after the bulk discount is applied. For a certain order with both the bulk discount and the coupon discount applied, the total price after the discounts for the sweaters purchased was 612 dollars. How many sweaters were purchased in this order?

A 19

B 20

C 30

D 46

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342 **Section 2, Module 1: Math**

[Meo]

$$x(kx - 48) = -8$$

In the given equation, k is an integer constant. If the equation has two distinct real solutions, what is the greatest possible value of k?

71

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341 **Section 2, Module 1: Math**

[Meo]

x	$t(x)$
27	14
36	$\frac{37}{2}$ **Meo**
49	25 **Meo**

For the linear function t, the table shows three values of x and their corresponding values of $t(x)$. Function v is defined by $v(x) = t(x) - 3$. What is the x-intercept of the graph of $y = v(x)$ in the xy-plane?

A $(6, 0)$

B $(5, 0)$

C $(2, 0)$

D $(-1, 0)$

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343 **Section 2, Module 1: Math**

[Meo] In triangle XYZ, the measure of angle X is 90 degrees. Point W lies on segment YZ, and segment WX is perpendicular to segment YZ. The length of segment WY is 548, and the length of segment WX is 411. What is the value of $\tan Z$?

A $\frac{4}{3}$

B $\frac{4}{5}$

C $\frac{3}{4}$

D $\frac{3}{5}$

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344 **Section 2, Module 1: Math**

[Meo] The mass of object A is 481% of the mass of object B, and the mass of object A is 0.740% of the mass of object C. If the mass of object C is $p\%$ of the mass of object B, what is the value of p?

65000

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345 **Section 2, Module 1: Math**

[Meo] In triangle ABC, the measure of angle B is 90 degrees and $\overline{BD}$ is an altitude of the triangle. The length of $\overline{AB}$ is 23 and the length of $\overline{AC}$ is 11 greater than the length of $\overline{AB}$. What is the value of $\frac{BC}{BD}$?

A $\frac{11}{23}$

B $\frac{23}{34}$

C $\frac{34}{23}$

D $\frac{23}{11}$

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347 **Section 2, Module 1: Math**

[Meo] If $5x - 8 = 7$, what is the value of $5x + 8$?

A -1

B 3

C 11

D 23

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346 **Section 2, Module 1: Math**

[Meo] The point $(9, 3)$ in the xy-plane is a solution to which of the following systems of inequalities?

A $x > 0$
$y > 0$

B $x > 0$
$y < 0$

C $x < 0$
$y > 0$

D $x < 0$
$y < 0$

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348 **Section 2, Module 1: Math**

[Meo] In the xy-plane, the graph of $y = \frac{1}{2}(x-2)(x+5)(x-c)$, where c is a constant, has a y-intercept of $(0, 40)$. What is the value of c?

A 4

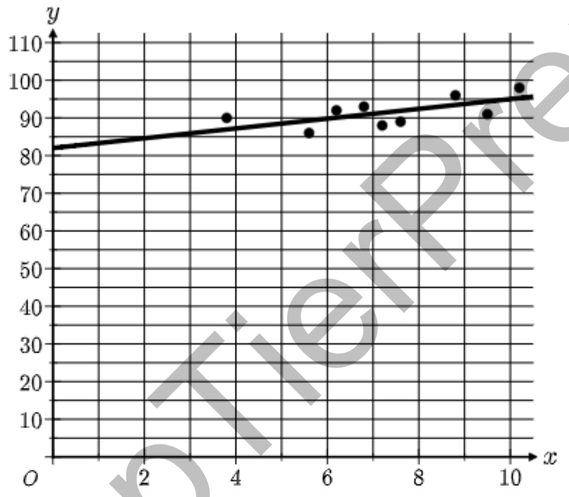
B 8

C 10

D 40

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349 **Section 2, Module 1: Math**



[Meo] The scatterplot shows the relationship between two variables, x and y, with 9 data points. A line of best fit for the data is also shown. For how many of the 9 data points is the actual y-value greater than the y-value predicted by the line of best fit?

A 8

B 7

C 5

D 2

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350 **Section 2, Module 1: Math**

[Meo] Triangle FGH is similar to triangle JKL, where angle F corresponds to angle J and angles G and K are right angles. If $\sin(F) = \frac{340}{389}$, what is the value of $\sin(J)$?

A $\frac{189}{389}$

B $\frac{340}{389}$

C $\frac{389}{340}$

D $\frac{389}{189}$

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351 **Section 2, Module 1: Math**

[Meo]

$$(7x^3 - 5) - (-2x^3 + 8)$$

The given expression is equivalent to $bx^3 - 13$, where b is a constant. What is the value of b?

9

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352 **Section 2, Module 1: Math**

[Meo] To install flooring, a contractor charges 7.75 dollars per square foot plus 100 dollars for labor. If a kitchen floor is a square with the side length of 15 feet, what is the total, in dollars, that the contractor charges to install flooring in this kitchen?

A 216.25

B 332.75

C $1,507.75$

D $1,843.75$

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354 **Section 2, Module 1: Math**

[Meo]

$$y = 13x + 6$$

The first equation in a system of two linear equations is given. The system has no solution. The second equation in the system can be written as $y = mx + b$, where m and b are constants. Which of the following CANNOT be the value of b?

A $\frac{1}{13}$

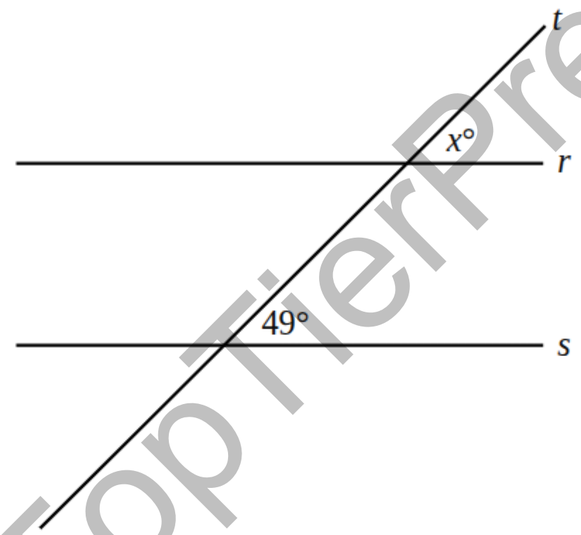
B $\frac{1}{6}$

C 6

D 13

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353 **Section 2, Module 1: Math**



Note: Figure not drawn to scale.

[Meo] In the figure shown, line r is parallel to line s. One of the angles formed by the transversal is 49 degrees, and another alternate interior angle is x degrees. What is the value of x?

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355 **Section 2, Module 1: Math**

[Meo]

Meo	Live north of Center St.	Live south of Center St. **Meo**	Total
Less than 45 years old	11	18	29 **Meo**
At least 45 years old	24	82	106 **Meo**
Total	35	100 **Meo**	135

The table summarizes members of a local organization by age and whether they live north or south of Center St. If a member of the organization is selected at random, what is the probability that the selected member is at least 45 years old?

A $\frac{29}{135}$

B $\frac{35}{135}$

C $\frac{100}{135}$

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356 **Section 2, Module 1: Math**

[Meo] The measure of angle Z is 120 degrees. What is the measure, in radians, of angle Z?

A $\frac{1}{3}\pi$

B $\frac{2}{3}\pi$

C $\frac{4}{3}\pi$

D 2π

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357 **Section 2, Module 1: Math**

[Meo]

$$40x + 36y = 720$$

Tom saved 40 dollars each month for x months, and Karla saved 36 dollars each month for y months. The given equation represents this situation, where Tom and Karla saved a total of 720 dollars. If Tom saved for 9 months, for how many months did Karla save?

A 40

B 20

C 18

D 10

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358 **Section 2, Module 1: Math**

[Meo] A length of 590 meters is equal to how many decimeters? (1 meter = 10 decimeters)

5900

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359 **Section 2, Module 1: Math**

[Meo] What is an x-coordinate of an x-intercept of the graph of $y = 3(x - 13)(x + 5)(x + 4)$ in the xy-plane?

13 -5 -4

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361 **Section 2, Module 1: Math**

[Meo] If $\frac{x}{y} = 3$ and $\frac{5x}{ty} = 105$, what is the value of t?

A $\frac{1}{7}$

B $\frac{3}{5}$

C $\frac{5}{3}$

D 7

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360 **Section 2, Module 1: Math**

[Meo] In the relationship between x and y, each increase of 1 in the value of x decreases the value of y by $\frac{4}{5}$. When $x = -5$, $y = -1$. Which equation represents this relationship?

A $y = -\frac{4}{5}x - 5$

B $y = -\frac{4}{5}x + 5$

C $y = \frac{4}{5}x - 5$

D $y = \frac{4}{5}x + 5$

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362 **Section 2, Module 1: Math**

[Meo]

$$y \geq x + 6$$

$$y \leq -3x - 6$$

Which point (x, y) is a solution to the given system of inequalities in the xy-plane?

A $(-4, 5)$

B $(-4, 1)$

C $(-2, 5)$

D $(-2, 1)$

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363 **Section 2, Module 1: Math**

[Meo] The function f is defined by $f(x) = -\frac{1}{2}(x - 11) + 6$. For what value of x does $f(x) = 3a$, where a is a constant?

A $-\frac{3a}{2} + \frac{1}{2}$

B $-\frac{3a}{2} + \frac{23}{2}$

C $-6a + 23$

D $-6a - 1$

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364 **Section 2, Module 1: Math**

[Meo] In a certain colony, one of the bees is a queen bee. Of the bees in this colony that are not the queen bee, 85% are worker bees, and the remaining bees are drones. If there are 11,004 drones in this colony, what is the total number of worker bees in this colony?

62356

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366 **Section 2, Module 1: Math**

[Meo] An exponential function h estimates the number of people in attendance at a festival x minutes after the start of the festival. The function estimates that there were 47 people in attendance at the start of the festival. The function also estimates that every 25 minutes for the first 150 minutes of the festival, the number of people in attendance increased by 133% of the number of people in attendance 25 minutes earlier. Which function defines h, where $x \leq 150$?

A $h(x) = 47(1.33)^{\frac{x}{25}}$

B $h(x) = 47(1.33)^{x+25}$

C $h(x) = 47(2.33)^{\frac{x}{25}}$

D $h(x) = 47(2.33)^{x+25}$

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365 **Section 2, Module 1: Math**

[Meo]

$$x - a = (x - a)(x - 35)$$

Which of the following are solutions to the given equation, where a is a constant and $a > 36$?

I. a
II. 35
III. 36

A I and II only

B I and III only

C II and III only

D I, II, and III

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367 **Section 2, Module 1: Math**

[Meo]

$$f(x) = 3^{-5(x+1)}$$

Which of the following equivalent forms of the given function f displays, as the base or the coefficient, the y-coordinate of the y-intercept of the graph of $y = f(x)$ in the xy-plane?

A $f(x) = \frac{1}{243}(\frac{1}{3})^{5x}$

B $f(x) = (\frac{1}{3})^{5x+5}$

C $f(x) = 3^{-5x-5}$

D $f(x) = 243^{-x-1}$

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368 **Section 2, Module 1: Math**

[Meo]

$$y = 2x^2 - 24x + 54$$

The given equation represents a parabola in the xy-plane. Which of the following equations that represent the same parabola displays the x-intercepts as constants or coefficients?

A $y = 2(x^2 - 12x) + 54$

B $y = 2x(x - 12) + 54$

C $y = 2(x - 6)^2 - 18$

D $y = 2(x - 9)(x - 3)$

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369 **Section 2, Module 1: Math**

[Meo] A right square pyramid has a surface area of $100 + 20\sqrt{554}$ square inches, which includes a base area of 100 square inches. What is the height, in inches, of this pyramid?

23

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370 **Section 2, Module 1: Math**

[Meo]

$$2x^2 - x\sqrt{k} + 2y^2 + y\sqrt{t} - 12 = 0$$

The given equation, where k and t are positive constants, defines a circle in the xy-plane. The radius of this circle is $\sqrt{57}$. Which expression represents the value of t?

A $180 - k$

B $k - 720$

C $720 - k$

D $816 - k$

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371 **Section 2, Module 1: Math**

[Meo]

	Live north of Center St.	Live south of Center St.	Total
Less than 45 years old	13 **Meo**	17	30
At least 45 years old	22	83 **Meo**	105
Total	35 **Meo**	100	135

The table summarizes members of a local organization by age and whether they live north or south of Center St. If a member of the organization is selected at random, what is the probability that the selected member is at least 45 years old?

A $\frac{30}{135}$

B $\frac{35}{135}$

C $\frac{100}{135}$

D $\frac{105}{135}$

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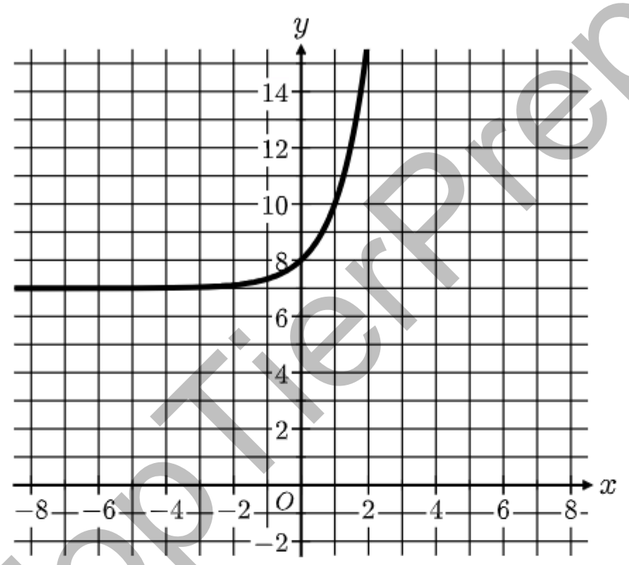
372 **Section 2, Module 1: Math**

[Meo] A chemist combines water and isopropanol to make a mixture with a volume of 31 milliliters (mL). The volume of isopropanol in the mixture is 10 mL. What is the volume of water, in mL, in the mixture? (Assume that the volume of the mixture is the sum of the volumes of water and isopropanol before they were mixed.)

21

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373 **Section 2, Module 1: Math**



[Meo] What is an equation of the graph shown?

A $y = 3^x + 7$

B $y = 3^{-x} + 7$

C $y = 3^x + 8$

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374 **Section 2, Module 1: Math**

[Meo] Cylinder A and cylinder B are each right circular cylinders with a radius of 3 inches. The height of cylinder A is half the height of cylinder B, and the total surface area of cylinder A is 102π square inches. If the cylinders are glued together along a circular base, what is the height, in inches, of the resulting cylinder?

(A) 6

(B) 14

(C) 28

(D) 42

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375 **Section 2, Module 1: Math**

[Meo] In the xy-plane, which of the following does NOT contain any points (x, y) that are solutions to $4x + 3y > 18$?

A The region where $x > 0$ and $y > 0$

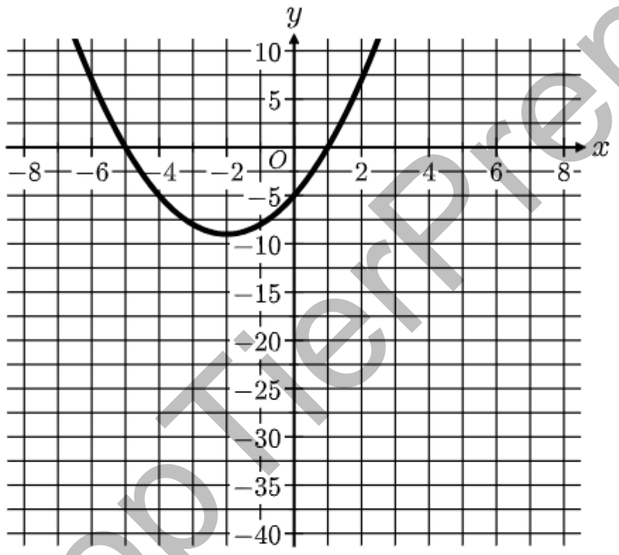
B The region where $x < 0$ and $y < 0$

C The region where $x > 0$ and $y < 0$

D The region where $x < 0$ and $y > 0$

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376 **Section 2, Module 1: Math**



[Meo] What is an equation of the graph shown, where $y = f(x)$?

A $f(x) = x^2 - 5$

B $f(x) = x^2 - 4x - 5$

C $f(x) = x^2 + 4x - 5$

D $f(x) = 2x^2 + 4x - 5$

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377 **Section 2, Module 1: Math**

[Meo] Triangle ABC is similar to triangle DEF, where angle A corresponds to angle D and angle C corresponds to angle F. Angles C and F are right angles. If $\tan(A) = \frac{25}{13}$, what is the value of $\tan(E)$?

13/25

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378 **Section 2, Module 1: Math**

[Meo] The function f is defined by $f(x) = \frac{15}{14-x}$. What is the value of $f(x)$ when $x = 9$?

3

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379 Section 2, Module 1: Math

[Meo]

x **Meo**	y **Meo**
0	-4 **Meo**
1	2
2 **Meo**	8 **Meo**

The table shows values of x and their corresponding values of y for three points on line j in the xy-plane. Line k also lies in the xy-plane and is defined by the equation $y = 5x$. At what point (x, y) do lines j and k intersect?

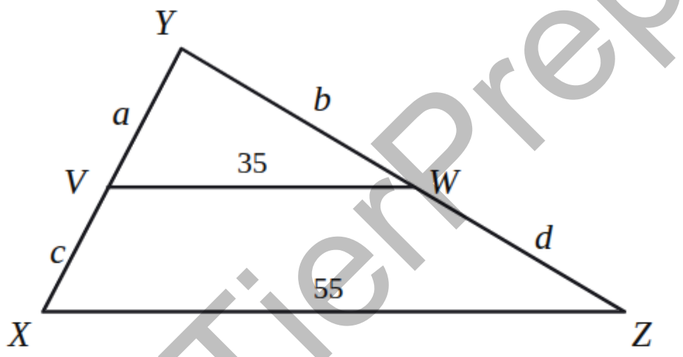
A $(4, 20)$

B $(4, -10)$

C $(5, 25)$

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380 **Section 2, Module 1: Math**



Note: Figure not drawn to scale.

[Meo] In triangle XYZ, a line segment VW is parallel to XZ, with $VW = 35$ and $XZ = 55$, as shown in the figure. If $YV = a$, $VX = c$, $YW = b$, and $WZ = d$, and we are given $\frac{a}{c} = k$ and $\frac{b}{d} = k$, what is the value of k?

A $\frac{4}{7}$

B $\frac{7}{4}$

C $\frac{7}{11}$

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381 **Section 2, Module 1: Math**

[Meo] The height of a right circular cylinder is 36 inches, and the circumference of its base is 360 inches. Which expression represents the total surface area, in square inches, of the cylinder?

A $(36)(360) + 2\pi(\frac{180}{\pi})^2$

B $(36)(360) + \pi(\frac{180}{\pi})^2$

C $\pi(\frac{180}{\pi})^2(36)$

D $(36)(360)$

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382 **Section 2, Module 1: Math**

[Meo]

$$x^2 + (\sqrt{k-3})x + 26 = 0$$

In the given equation, k is a constant. The equation has exactly one real solution. What is the value of k?

A 29

B 101

C 104

D 107

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384 **Section 2, Module 1: Math**

[Meo]

$$\frac{3}{c} - \frac{5}{d} = 7$$

The given equation relates the variables c and d, where c and d are greater than 1. Which equation correctly expresses d in terms of c?

A $d = \frac{c}{5(3-7c)}$

B $d = \frac{5c}{3-7c}$

C $d = \frac{5(3-7c)}{c}$

D $d = \frac{3-7c}{5c}$

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383 **Section 2, Module 1: Math**

[Meo] When several resistors are connected in series in a circuit, the total resistance of the circuit is the sum of the resistances of these resistors. A certain circuit consists of 8 resistors connected in series, where the resistance of each resistor is positive. The total resistance of 3 resistors in this circuit is 80 ohms. Each of the other 5 resistors in this circuit has a resistance of at most 20 ohms. The total resistance of this circuit is x ohms. Which inequality best represents this situation?

A $0 < x \leq 180$

B $80 < x \leq 100$

C $80 < x \leq 180$

D $x \geq 100$

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385 **Section 2, Module 1: Math**

[Meo]

$$f(x) = 11^{-4(x+1)}$$

Which of the following equivalent forms of the given function f displays, as the base or the coefficient, the y-coordinate of the y-intercept of the graph of $y = f(x)$ in the xy-plane?

A $f(x) = \frac{1}{14,641}(\frac{1}{11})^{4x}$

B $f(x) = (\frac{1}{11})^{4x+4}$

C $f(x) = 11^{-4x-4}$

D $f(x) = 14,641^{-x-1}$

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386 **Section 2, Module 1: Math**

[Meo] Which of the following expressions has a factor of $x + 2b$, where b is a positive integer constant?

A $3x^2 + 5x + 10b$

B $3x^2 + 26x + 10b$

C $3x^2 + 42x + 10b$

D $3x^2 + 47x + 10b$

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387 **Section 2, Module 1: Math**

[Meo] Line c is defined by $y = 5x + 8$. Line d is parallel to line c in the xy-plane. What is the slope of line d?

A 3

B 5

C 8

D 13

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388 **Section 2, Module 1: Math**

[Meo] Which expression is equivalent to $7xy + 35x^2y + 42xy^2$?

A $7xy(1 + 35x + 42y)$

B $7xy(1 + 5x + 6y)$

C $7xy(35x + 42y)$

D $7xy(5x + 6y)$

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390 **Section 2, Module 1: Math**

[Meo]

$$3(x + \frac{b}{3} - 28) = 3x$$

In the given equation, b is a constant. The equation has infinitely many solutions. What is the value of b?

84

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389 **Section 2, Module 1: Math**

[Meo] If $|14 - 2x| = 42$, which of the following is a value of $7 - x$?

A 0

B 7

C 21

D 28

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391 **Section 2, Module 1: Math**

[Meo] The expression $9(x-3)(x+3)(x^2-4)$ is equivalent to $9x^4-rx^2+p$, where r and p are constants. What is the value of $p-r$?

A 207

B 5

C -5

D -207

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392 **Section 2, Module 1: Math**

[Meo]

$$f(x) = 9((\frac{1}{8})^x + 4)$$

The function f is defined by the given equation. For all values of x, the value of $f(x)$ is greater than k, where k is a constant. What is the greatest possible value of k?

A 0

B 4

C 36

D 288

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394 **Section 2, Module 1: Math**

[Meo] As the size of a firework's shell increases, the height above the ground when the firework bursts increases. For a firework to burst 858 feet above the ground, a firework's shell in the shape of a sphere with a volume of 282.40 cubic inches could be used. To the nearest whole number, what is the volume of this firework's shell in cubic centimeters? (1 inch = 2.54 centimeters)

4628

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393 **Section 2, Module 1: Math**

[Meo]

$$y > 7x + k$$

$$y < 9 - x$$

In the given system of inequalities, k is a positive constant. Which of the following represents the x-coordinates of all the points (x, y) that satisfy this system in the xy-plane?

A $x > \frac{k}{9}$

B $x < \frac{k}{9}$

C $x > \frac{9-k}{8}$

D $x < \frac{9-k}{8}$

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395 **Section 2, Module 1: Math**

[Meo] For two acute angles X and Y, the measure of angle Y is 28 degrees and $\frac{\sin X}{\cos Y} = 1$. What is the measure, in degrees, of angle X?

A 28

B 45

C 62

D 73

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396 **Section 2, Module 1: Math**

[Meo] A parabola in the xy-plane has a vertex of $(1, 20)$. The equation $y = -ax^2 + bx + c$ represents this parabola, where a, b, and c are positive integer constants. What is the greatest possible value of b?

38

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397 **Section 2, Module 1: Math**

[Meo] At a nature preserve, a wildlife biologist counted bald eagles from an observation deck at the same time each day for 21 days.

Number of bald eagles	Number of days **Meo**
0	1
1	3
2	4
3	5
4	4
5	3
19	1

The table summarizes the resulting data set, data set A. The data value 19 was recorded in error and is removed from data set A to create data set B, which consists of the remaining 20 data values. Which statement best compares the median of data set A and the median of data set B?

A The median of data set B is less than the median of data set A.

B The median of data set B is equal to the median of data set A.

C The median of data set B is greater than the median of data set A.

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398 **Section 2, Module 1: Math**

[Meo] Which of the following expressions has a factor of $x + 2b$, where b is a positive integer constant?

A $3x^2 + 5x + 10b$

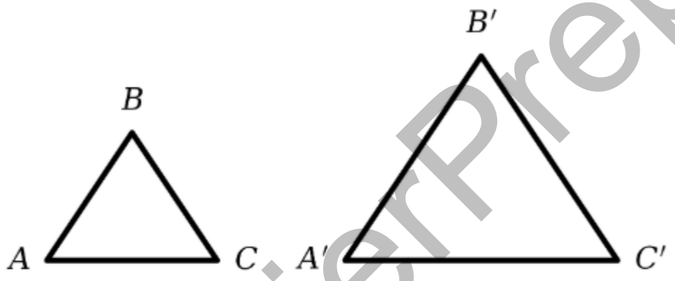
B $3x^2 + 26x + 10b$

C $3x^2 + 42x + 10b$

D $3x^2 + 47x + 10b$

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399 **Section 2, Module 1: Math**



Note: Figures not drawn to scale.

[Meo] Triangles ABC and $A'B'C'$ are shown. Triangle ABC is dilated by a scale factor of 6 to form triangle $A'B'C'$. If the length of $\overline{AB}$ is 18, what is the length of $\overline{A'B'}$?

A 3

B 6

C 24

D 108

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401 **Section 2, Module 1: Math**

[Meo]

$$7x^2(5x-17)(5x-u)=0$$

In the given equation, u is a positive constant. The sum of the solutions to the equation is 8. What is the value of u?

A 8

B 17

C 23

D 39

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400 **Section 2, Module 1: Math**

[Meo] The expression $13x(x^2 + 6)$ is equivalent to $ax^b + cx$, where a, b, and c are constants. What is the value of abc?

3042

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402 **Section 2, Module 1: Math**

[Meo] Cylinder A and cylinder B are each right circular cylinders with a radius of 3 inches. The height of cylinder A is half the height of cylinder B, and the total surface area of cylinder A is 96π square inches. If the cylinders are glued together along a circular base, what is the height, in inches, of the resulting cylinder?

A 6

B 13

C 26

D 39

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403 **Section 2, Module 1: Math**

[Meo]

$$3(x + \frac{b}{3} - 17) = 3x$$

In the given equation, b is a constant. The equation has infinitely many solutions. What is the value of b?

51

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404 Section 2, Module 1: Math

[Meo]

$$r + g < 39$$

For which of the following tables are all the values of r and their corresponding values of g solutions to the given inequality?

A

r **Meo**	g
0	34 **Meo**
2	36 **Meo**
4	38

B

r	g
0	34
2 **Meo**	38
4	38 **Meo**

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0	34
2 **Meo**	38
4	36 **Meo**

C

r	g
0	36
2 **Meo**	38 **Meo**
4	34

D

r **Meo**	g **Meo**
0	38
2	36
4	34

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405 **Section 2, Module 1: Math**

[Meo] If $2\sqrt{6x} = 4$, what is the value of $30x$?

A 10

B 20

C 40

D 60

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406 **Section 2, Module 1: Math**

[Meo] During a physics experiment, the speed of an object was determined to be 145.0 inches per second. What is this speed in feet per second? (12 inches = 1 foot)

145/12

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407 **Section 2, Module 1: Math**

[Meo]

$$y = 30x + 6$$

Which table gives three values of x and their corresponding values of y for the given equation?

A

x	y
0 **Meo**	30
2	60
4 **Meo**	120 **Meo**

B

x **Meo**	y
0	6
2	66
4 **Meo**	126

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0	6
2	66
4 **Meo**	126

C

x	y
0 **Meo**	6
2	54
4	114

D

x **Meo**	y
0	30
2	36
4 **Meo**	42 **Meo**

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408 **Section 2, Module 1: Math**

[Meo]

$$g(x) = 11(\frac{1}{7})^x$$

If the given function g is graphed in the xy-plane, where $y = g(x)$, what is the y-intercept of the graph?

A $(0, 11)$

B $(0, 7)$

C $(0, 77)$

D $(0, 4)$

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409 **Section 2, Module 1: Math**

[Meo] A function p estimates that there were $2,500$ whales in a population in 1998. Each year from 1998 to 2010, the function estimates that the number of whales in this population increased by 7% of the number of whales in the population the previous year. Which equation defines this function, where $p(x)$ is the estimated number of whales in the population x years after 1998?

A $p(x) = 2,500(7)^x$

B $p(x) = 2,500(1.93)^x$

C $p(x) = 2,500(1.07)^x$

D $p(x) = 2,500(0.93)^x$

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410 Section 2, Module 1: Math

[Meo]

x **Meo**	y
-5	$91 + 4k$ **Meo**
5 **Meo**	91
10	$91 - 2k$

The table gives three values of x and their corresponding values of y, where k is a positive constant. There is a linear relationship between x and y. Which equation represents this relationship?

A $2kx - 5y = 91 + 2k$

B $2kx - 5y = 455 + 10k$

C $2kx + 5y = 91 + 2k$

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411 **Section 2, Module 1: Math**

[Meo] A marine biologist designed an aquarium for dolphins in the form of a right cylinder with a depth of 12 feet and a volume of 145, 800 cubic feet. What is the volume of this aquarium, in cubic yards? (1 yard = 3 feet)

5400

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413 **Section 2, Module 1: Math**

[Meo] A cell phone plan costs 83 dollars each month. Which equation gives the total cost c, in dollars, of the cell phone plan for m months?

A $c = \frac{83}{m}$

B $c = m - 83$

C $c = 83m$

D $c = m + 83$

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412 **Section 2, Module 1: Math**

[Meo] If $x = \sqrt[3]{2x^n + 75}$, where n is a positive integer constant, what is the value of x^n?

25

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414 Section 2, Module 1: Math

[Meo]

$$5x - \frac{1}{10}y = 246$$

$$\frac{1}{5}x + y = 50$$

The solution to the given system of equations is (x, y). What is the value of x?

A 5

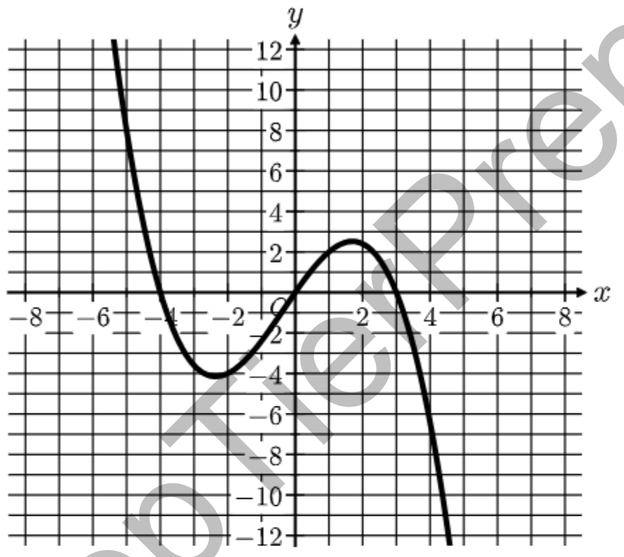
B 15

C 50

D 55

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415 **Section 2, Module 1: Math**



[Meo] For the polynomial function f, the graph of $y = f(x)$ is shown. Which of the following must be a factor of $f(x)$?

A $x - 4$

B $x - 3$

C $x + 3$

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416 **Section 2, Module 1: Math**

[Meo]

$$(d-33)(d+33)-9=-9$$

What is a solution to the given equation?

33 -33

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417 **Section 2, Module 1: Math**

[Meo] In the xy-plane, which of the following does NOT contain any points (x, y) that are solutions to $4x + 3y > 15$?

A The region where $x > 0$ and $y > 0$

B The region where $x < 0$ and $y < 0$

C The region where $x > 0$ and $y < 0$

D The region where $x < 0$ and $y > 0$

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418 **Section 2, Module 1: Math**

[Meo] Triangles ABC and DEF are similar, where A corresponds to D and B corresponds to E. The measure of angle F is $65°$ and $AB = \frac{1}{5}DE$. Which of the following must be true?

I. The measure of angle C is $13°$.
II. $5BC = EF$.

A I only

B II only

C I and II

D Neither I nor II

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419 **Section 2, Module 1: Math**

[Meo]

$$\frac{1}{42}x = \frac{1}{84}x$$

How many solutions does the given equation have?

A Zero

B Exactly one

C Exactly two

D Infinitely many

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420 **Section 2, Module 1: Math**

[Meo] Which expression is a factor of $x^4 + 14ax^2 + 49a^2 - 64$, where a is a positive constant?

(A) $x^2 + 7a + 8$

(B) $x^2 - 7a - 8$

(C) $x^2 + 14a$

(D) $x^2 - 8a$

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421 **Section 2, Module 1: Math**

[Meo] For $t > 0$, the function s is defined as follows:

$$s(t) \text{ equals } t \text{ less than } 74\% \text{ of } t$$

Which of the following could describe this function?

A Decreasing linear

B Decreasing exponential

C Increasing linear

D Increasing exponential

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422 **Section 2, Module 1: Math**

[Meo] The estimated total pressure a diver experiences below the surface of the water is equal to the sum of the estimated water pressure and the estimated atmospheric pressure. At a depth of h meters below the surface, the estimated total pressure P_1, in kilopascals (kPa), a diver experiences is given by $P_1 = 10h + 101$. After maintaining a depth of h meters, the diver descended d additional meters. The estimated total pressure P_2, in kPa, the diver experienced after descending these d additional meters is given by $P_2 = 10d + 341$. What is the estimated total pressure, in kPa, the diver experienced at a depth of h meters below the surface?

A 442

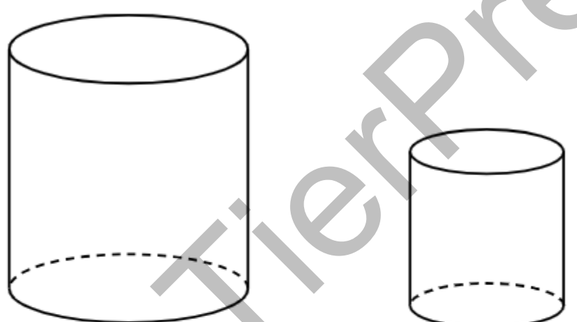
B 341

C 101

D 10

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423 **Section 2, Module 1: Math**



Note: Figure not drawn to scale.

[Meo] Cylinder A is similar to cylinder B. The volume of cylinder A is 5.13 cubic units and the volume of cylinder B is 1.52 cubic units. The surface area of cylinder A is 16.47 square units. What is the surface area, in square units, of cylinder B?

7.32

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424 **Section 2, Module 1: Math**

[Meo] A parabola in the xy-plane has a vertex of $(1, 35)$. The equation $y = -ax^2 + bx + c$ represents this parabola, where a, b, and c are positive integer constants. What is the greatest possible value of b?

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